

Netso Developer Handoff: Technical Setup & API Guide

1. System Overview

Netso App is a mobile-first energy intelligence platform for rooftop solar users in Bangladesh. It pulls real-time inverter data, calculates savings from net metering, and enables services like panel cleaning and referrals.

2. Tech Stack

- **Frontend:** React Native (Expo)
- **Backend:** Node.js + Express
- **Database:** MongoDB Atlas
- **Auth:** Firebase OTP / Google Login
- **Push Notifications:** Firebase Cloud Messaging (FCM)
- **IoT Inverter Data:** MQTT broker (Deye / Growatt)
- **Payments:** Stripe for global / bKash + Nagad for local
- **Deployment:** Firebase + GitHub Actions CI/CD

3. App Modules

Module	Path	Component
Dashboard	<code>/dashboard</code>	SolarDataView, SavingsCard
Earnings	<code>/finance</code>	PaybackGraph, ROIProgress
Referrals	<code>/referrals</code>	BonusTracker, InviteLink
Maintenance	<code>/services</code>	ServiceRequestForm
Notifications	Global context	AlertBadge, SummaryCard

4. Auth & User

- Firebase Auth (Phone login w/ OTP)
- Google SSO optional
- User data stored in MongoDB (`users` collection):
 - name, phone, email
 - install_address
 - inverter_serial, site_id



5. Inverter Data Flow

1. Inverter posts real-time data via MQTT to our broker
2. Broker triggers backend `/api/inverter/push`
3. Data is parsed and written to `site_data` collection
4. React Native fetches via `/api/user/:id/dashboard` (every 5 min)



6. Net Metering Earnings API

- `/api/export/month/:userid`
- Calculates $(\text{units_exported} \times \text{slab_rate})$
- Uses latest DESCO/DPDC rate chart (JSON-config)
- Returns:
 - `monthly_earning` (BDT)
 - `export_kWh`
 - `CO2 offset equivalent`



7. Notification Triggers

- Low generation / inverter offline
- Upcoming service schedule
- Monthly report ready

Backed by FCM → triggered from Node backend (`/alerts/job`) or CRON job



8. Testing / QA Notes

- Use simulated inverter JSON data for local dev
- Battery fallback logic should simulate 5 min loss intervals
- UI dark mode tested on Android v11+ / iOS 14+



9. Deployment & CI/CD

- GitHub repo: [project-link]
- Push to `main` triggers CI → Expo OTA + Firebase Hosting update
- Environment vars in `.env.local`:
 - `MONGODB_URI`
 - `FIREBASE_KEY`
 - `MQTT_USER` / `MQTT_PASS`
 - `STRIPE_KEY`

10. API Summary

- `POST /api/login` - OTP auth
- `GET /api/dashboard/:id` - user solar + financials
- `POST /api/referral/:id` - track leads
- `GET /api/services/:id` - maintenance options
- `POST /api/inverter/push` - receive MQTT live packet

This handoff document includes all core modules, endpoints, and infrastructure for launching and scaling the Netso App v1.0 with real-time solar intelligence for Bangladesh.